

## PATIENT

**Raul Falcon**

DOB 10/18/2002

AGE 22 yrs

SEX Male

PRN F 430575

## FACILITY

**Functional Mind LLC**

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## ENCOUNTER

**Telemedicine Visit**

NOTE TYPE SOAP Note

SEEN BY Achina Stein

DATE 07/03/2025

AGE AT DOS 22 yrs

Not signed

**Plan**

Hi Raul,

It was good to see you and your parents again. Your MSQ has dropped down to 8 from 27 which is significant! It sounds like stressors may have decreased (end of exams, etc.) which would have helped. It is also important recognize other interventions that may be contributing to this improvement - specifically the B6 complex and the dietary changes you have made. Testing revealed the severe deficiency in a few b vitamins and this correlates with mental health and mood disorders. In addition, removing gluten, minimizing starches/carbs, and eating more good fats - even if not in ketosis - will lower gut and brain inflammation tremendously and contribute to positive shifts. You will need to look at your diet as a medical intervention. This is making a conscious choice about your mental health, that is if the goal is ultimately to taper and discontinue medication. I'd like you to commit to at least 6 months of this intervention. Let all of your family and friends know that you need their support in making this happen.

We like to track all changes and interventions so that we can refer to this in the future as a "tool box" of what items "move the needle."

The functional labs revealed several healing opportunities which is where we will put focus the next few months. By eliminating stressors internally and restoring your gut, many more positive shifts can be made. I am providing 3 months of a gut restoration plan specific to you. After that we will repeat some testing and consider at that a time a medication change.

This is your optimal health plan and D.R.E.S.S.® Protocol. DRESS stands for diet, rest, exercise, stress, and supplements. Any hyperlinks will need to be copied and pasted.

**ACTION ITEMS:**

- Please upload your most recent Lithium level and other labs not already sent
- **Watch the digestion video: <https://vimeo.com/431544909/e121557744>**
  - PW for all videos: achinastein2023
- Make another appointment with Nancy, your health coach.
- Read through your new plan and ask questions as needed

**LAB RESULTS - Overview/Highlights**

## GI Map

- Commensal bacteria - Low diversity with some missing species - major contributors to a dysfunctional immune system in the gut
- Mild Dysbiosis - opportunistic bacteria
- Elastase suboptimal - which is a marker for pancreatic enzyme output. This should be well over 500 for your age. The burp test was failed so there may not be enough HCl
- Secretory IgA EXTREMELY HIGH - immune response/activation at the gut lining
- Anti-Gliadin EXTREMELY HIGH - Our cutoff is ~80. Your level is indicative of non-celiac gluten sensitivity meaning you are highly gluten sensitive
  - There are other markers for gluten sensitivity/intolerance/allergy; this is one
  - You will need to take gluten fully out of your diet
- Calprotectin - mild/moderate inflammation noted; this is a marker for inflammation in the colon which may be due to the gluten. We will repeat calprotectin at a later date

## OAT

- Mild amount of arabinose which is a marker for Candida, your Candida score on the Intake was 116 which indicates that Candida is a highly probable pathogen and this matches your cravings for carbs as you move to the ketogenic diet.

- Mitochondrial dysfunction - Carnitine and coQ10, B2, will support mitochondrial and keto diet
- Neurotransmitters are affected by medications so this is a non clue
- Deficient in B2, B6, C; insufficient in B5
  - Your serum B12 was deficient which is different than urine testing. The MMA indicates that it is sufficiently utilized in the body

#### Mycotoxins

- 11.36 Ochratoxin A
  - There may be more mycotoxins in the body but we may not see it until we pull them out with binders
- Citrinin is mildly elevated and may increase when pulled out on way to excretion

#### TOXDetect

- Negative for most toxins

#### Toxic Metals

- Arsenic and mercury are the most elevated but in normal ranges
- Binders will help pull these out

#### Glyphosate

- Elevated - which can cause mood disorders and is neurotoxic; binders will pull this out

#### Bloodwork

- Nutrients - many insufficiencies and deficiencies - optimizing these will help overall function
  - Serum B12 is lab low (optimal at 1000); can be root cause for depression, psychosis, and other psychiatric conditions all by itself. This level is even lower than when your psychiatrist did the bloodwork
  - Folate - suboptimal (optimal is >20)
  - Vitamin D 42.2 - this is suboptimal. The goal is 60-80
  - Vitamin A insufficient (goal 125)
  - Zinc: copper ratio is imbalance slightly - we want a 1:1 ratio
- Inflammatory Markers
  - Homocysteine 9 - this has to do with how your body utilizes B12 and folate
    - Optimally this would be 6-8
  - hsCRP - WNL; Sed Rate WNL
- Cholesterol panel
  - Suboptimal HDL (goal >60); the Keto diet may help but fish oil can improve
- Glycemic Control
  - Insulin, glucose and A1C are WNL
- WBC's - low normal which is indicative of chronic infection
  - Neutrophils - mildly elevated indicating some bacterial infection/colonization
  - Granulocytes are another clue for bacterial colonization
  - Monocytes, eosinophils and basophils are activated but not high - it can point to parasitic activity may be present but hidden by biofilms
- Fatty Acid profile - imbalanced
  - You will benefit from more omega 3 foods and supplementation
  - Omega 3's are anti-inflammatory and omega 6's are
- MTHFR - 1298 - heterozygous and 677 wild homozygous -some issues utilizing folate and B12 at the level of the brain which means you should only use methylated vitamins, the Phytomulti has methylated B12 and methyl folate. Never take any synthetic B vitamins.

#### DIET:

I know it is difficult to follow a keto diet if you are trying to source ingredients or going out to eat. Even so, getting more good fats in your diet by following it to about 70% is a great start.

There is a continuum as it relates to diet - not everyone needs to achieve ketosis to get benefits from adding good fats and removing sugars and starches from their diet. Be sure to celebrate the success you have had thus far. We look for progress, not perfection.

#### Recommendations:

- **Track ketones daily** at the start to help you find your "sweet spot" as you continue to work toward ketosis. The meter you have is great to track ketones and glucose.
- **You MUST remove gluten from your diet** as you have non-celiac gluten sensitivity. This is a major root cause of mental health problems.
  - If you have not already, you might consider reviewing the gluten information from the previous plan so you can find the **hidden sources of gluten** that can show up (even when eating a keto diet)
  - Examples: soy sauce, dressings, condiments, species or seasonings
  - Examples: keto baked goods, maltodextrin as an additive, food starch
- If there are ingredients you do not feel will support you in ketosis or do not resonate with you, feel free to reach out to April to discuss, or just omit these when trying a **FAT BOMB** recipe.
- Try and **eat foods higher in zinc and omega 3** fatty acids:

#### Sources of Zinc in Food

- Rock oysters, Crab, Lobster, sardines
- Lamb shank, Beef (steak), Chicken breast
- Pumpkin seeds (pepitas), Almonds, Cashews
- There are others that are not keto-friendly not listed here

#### List of Foods Highest in Omega 3 Fatty Acids

- Flax seed, walnuts, chia seeds
- Small or large high quality oily fish like sardines, mackerel, herring, oysters, or salmon
- Organic edamame and seaweed

#### Hydration

- Aim for half your weight in ounces, but have your hydration be consistent daily so that your body can acclimate and Lithium levels are not affected since they are processed via the kidneys.
- We DO NOT recommend electrolytes while you are on Lithium

#### REST:

- I am glad you are falling asleep more quickly and feel rested when you wake up
- It sounds like 9 hours is your baseline
- I am glad you have been able to taper the sleep medications and have also altered your bedtime to a later bedtime.
- I now the goal is to eventually remove this medication - Prosom
- Wake up in the morning to get first light and walk early in the morning in the sun to get your Vitamin D level higher without sun glasses and sunscreen for 20 minutes

#### EXERCISE:

- I am glad exercise has been more consistent and helps your mental health!
- Continue walking - it sounds like you are walking more than previously
- Continue strength training 2-4X weekly
- Movement is vital for mental health and moving lymph

#### STRESS REDUCTION:

Though we did not get to discuss it, it sounds like you have made strides to mitigate stress in your life which will support digestion and healing.

- Please watch the digestion video (above)
- Continue to work with Nancy on a stress reduction practice that resonates with you

#### SUPPLEMENTS:

Always sequence and titrate new supplements:

- Sequence - start each supplement one at a time for at least 2 days
- Titrate - some supplements will need to be increased gradually.

How to do this:

- Add them all in one at a time at the lowest dose and then go back to the beginning and increase dosing as tolerated until you reach targeted dosing recommendation

### Finish and Discontinue

- Ortho Digestzymes - take 1 capsule at the start of each meal to enhance digestion
- If digestion gets worse, I have provided Betaine HCl only to digest proteins since you failed the BURP test but you may not need that if you work to practice the techniques noted in the Digestion video above.

### Continue Foundational Supplements:

- Metagenics PhytoMulti - take 2 capsules daily with food
  - This is a highly bioavailable multi to help fill the wells of nutrient deficiencies
- Vital Nutrients B6 + B Complex - take 1 capsule daily with food - to improve your homocysteine
- Seeking Health Vitamin D3 K2 5000 - take 1 capsule daily with food

### Start

- Zinc Picolinate 15 mg - 1 capsule daily with food for 3 months to flip the zinc/copper ratio and assist with B vitamin utilization. Zinc is a cofactor in many reactions and lower copper levels which will help to remove Candida. Candida loves a high copper environment.
- Nordic Naturals ProOmega 2000 - 2 gel caps daily with food to increase HDL and improve omega3/6 ratio
- Vital Nutrients carnitine 500 mg 2X daily with food for mitochondrial support
- Life Extension Super Ubiquinol CoQ10 with PQQ - 1 gel cap daily with food for mitochondrial support

### THEN Start Gut Restoration

**\* You will stay on Foundational Supplement above throughout and switch supplements month to month below**

**\* We recognize this is a lot of capsules but is time-limited**

### MONTH 1 Kill Phase for 1 MONTH

- **Apex Energetics GI Synergy** 1 packet 3X daily
  - this is for pathogens like candida, opportunistic bacteria and potential parasites
- **GI Detox** - 3 capsules 1X daily in the AM on a FULLY EMPTY STOMACH (2 hours away from food, medications, and supplements)
  - **START and Titrate up from 1 capsule daily until you can tolerate 3**
  - You may need this for a maximum of 6 months to bind to mold, metals, glyphosate and other environmental toxins
  - **Please make sure you are having daily BMs** as this supplement can be constipating.
  - Take **cascara sagrada** 1-3 capsules at bedtime, if this happens.
  - Please make sure you are not living in a mold environment. If you are, this must be remediated

### Repair and Re-inoculation Phase

#### MONTH 2

- Ortho Molecular Orthobiotic - 1 capsule every night
  - This is a probiotic - 20 billion CFUs with Sacch B and does not need to be refrigerated
- Megasporebiotic - 1 capsule twice daily -am and bedtime. This is a spore based probiotic that heals the gut lining.
  - Take for 2-3 months total.
- Apex Energetics Enterovite - Take 1 capsule 2 times daily. Take for 2-3 months total.
  - It provides short chain fatty acids and helps to eliminate the food sensitivities.
- Apex Energetics RepairVite GT - 1 scoop two times a day initially preferably on an empty stomach.
  - This is to repair the gut lining and bring the inflammation. Take for 2 -3 months total.
  - This will help calm the gut and bring down the Secretory IgA
- Pendulum Akkermansia 100mg- take 1 capsule daily with food

#### MONTH 3

- DFH ProbioMed 50- 50 billion CFU at bedtime for 1 month. This is a higher dose broad spectrum probiotic.
- Apex Energetics Enterovite - Take 1 capsule 2 times daily.
- Megasporebiotic - 1 capsule twice daily -am and bedtime.
- Apex Energetics RepairVite GT - 1 scoop three times a day or 1.5 scoops twice daily.
- Pendulum Akkermansia 100mg - take 1 capsule daily with food

## MONTH 4

- DFH ProbioMed 100- 100 billion CFU at bedtime for 1 month. This is a higher dose broad spectrum probiotic.

## Finish and discontinue:

- Apex Energetics Enterovite - Take 1 capsule 2 times daily.
- Megasporebiotic - 1 capsule twice daily -am and bedtime.
- Apex Energetics RepairVite GT - 1 scoop three times a day or 1.5 scoops twice daily.
- Pendulum Akkermansia 100mg - take 1 capsule daily with food

**RESOURCES:****Gluten Sensitivity Presenting as a Neuropsychiatric Disorder**

- <https://pmc.ncbi.nlm.nih.gov/articles/PMC3944951/>

**Gut inflammation and increased gut permeability leads to autoimmunity**

- <https://www.ncbi.nlm.nih.gov/pmc/articles/PMC7767453/>

**REGARDING KETO**

- If you need assistance with food, please reach out to set a coaching appt with April
- You may find it helpful to join a keto group on FB or follow people on Insta for support and information

**From the last plan, here are some resources I will move forward:**

- Dr Georgia Ede Website:
- <https://www.diagnosisdiet.com/>
- Keto Guide from Metagenics will be uploaded to IntakeQ as an additional resource, but please know you:
- Want to follow a dairy free version
- Can use recipes from Dr Ede's book or other resources
- Watch videos on You Tube on channel - Metabolic Mind

There are also many Facebook Groups that can support you on this journey into keto.

**FOLLOW UP appointment: September 24, 2025 at 11:00 am ET in the OFFICE.**

**Please check your schedule to make sure you can fit this time in. The only other option is Zoom sessions on Monday and Tuesdays.**

A link will be sent to you via email with the pre-appt form ahead of time.

You will receive a pre-appt form to complete 10 days before the appt. Please complete 48 hours or earlier than the appt to give time to get it uploaded into the computer. We don't want to take time during the appt to do this for my review

Let me know if you have any questions.

Achina Stein DO IFMCP  
April Ritchotte FDN-P

Total time spent face to face: 90 min. Total time spent counseling/education was greater than 50%Content of counseling: Educated about nutrition, reason for dietary changes, gut permeability and its causes, stress management, importance of exercise, sleep hygiene.

Medications/Prescription orders attached to encounter:

Lamotrigine (lamoTRigine) 100 MG Oral Tablet Sig: Take 1 tablet by mouth daily